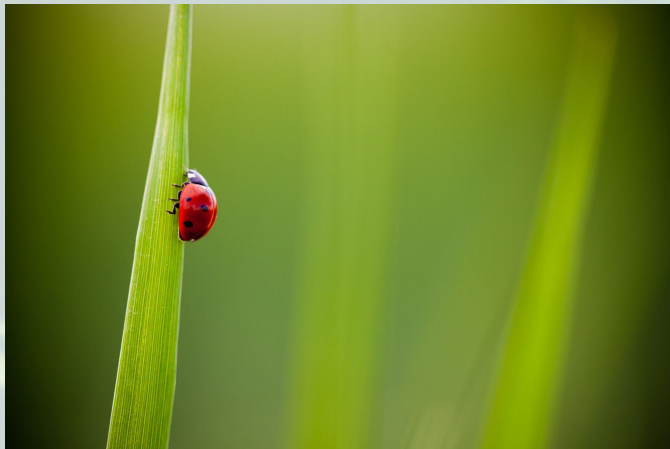




Vascular & Non-vascular Plants

What types of plants are there?



What is Vascular?

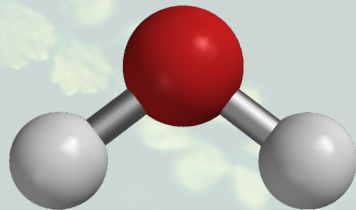
Vascular- having tubes that move fluids through a body



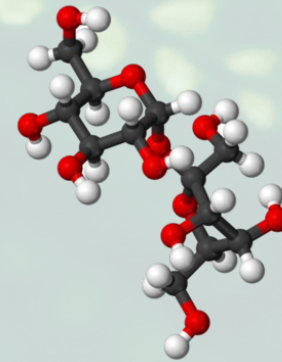
Are humans vascular?

Yes! We have veins and arteries so our bodies can transport oxygen using red blood cells.

What materials would plants need to transport through their bodies?

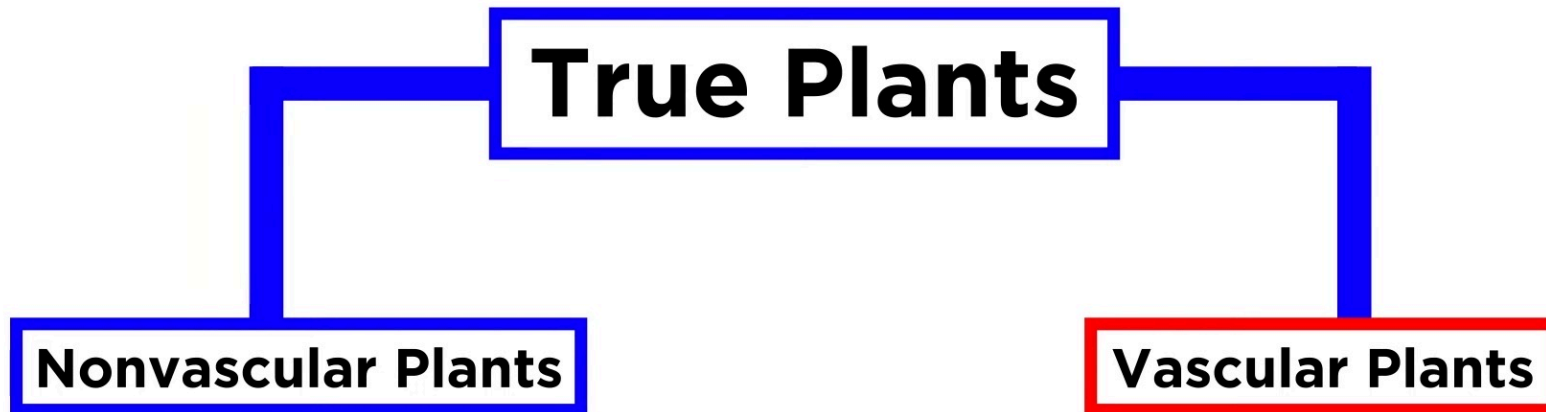


Water!
Carried by
Xylem



Sugar!
Carried by
Phloem

What types of plants are there?



do not contain
xylem and **phloem**



do contain
xylem and **phloem**

Non-Vascular Plants (Bryophytes)

- The first types of land plants to separate from algae
- Do not have special tissues to transport water and nutrients, including xylem and phloem
- No long stems or complicated structures
- Have rhizoids instead of roots
- Need water to reproduce. They cannot spread spores on the wind or scatter seeds

Examples of Non-Vascular Plants



Liverwort



Hornwort



Moss

Vascular Plants (Tracheophytes)

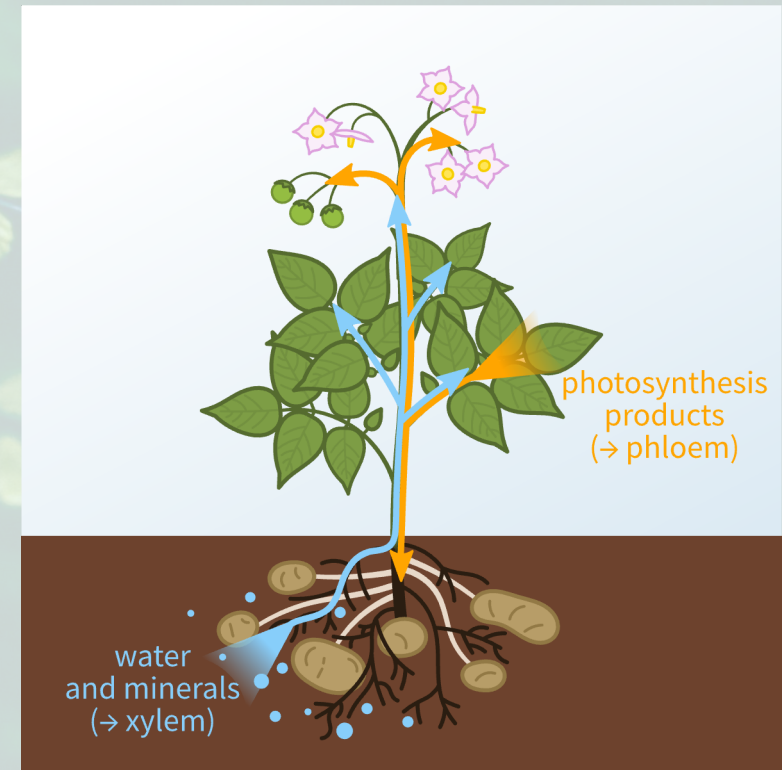
have special tissue that move water and nutrients through the plant.

Xylem

transports water and minerals from the roots to the stalks and leaves

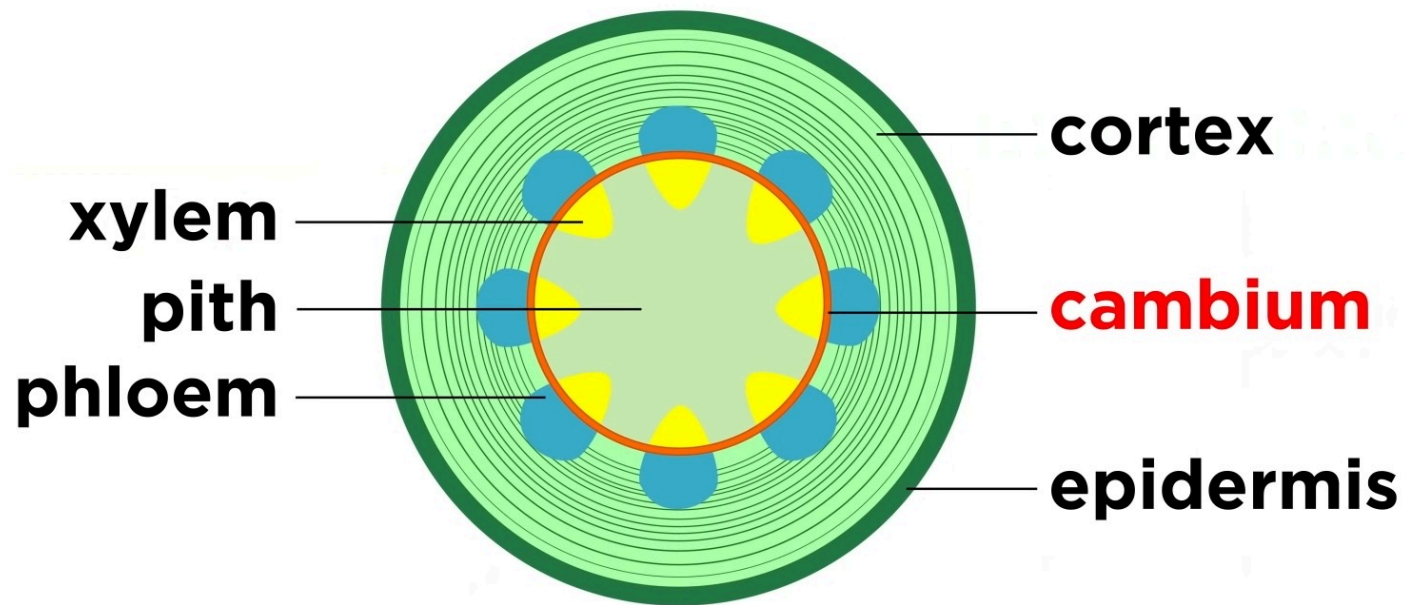
Phloem

transports sugars made in photosynthesis to the rest of the plant



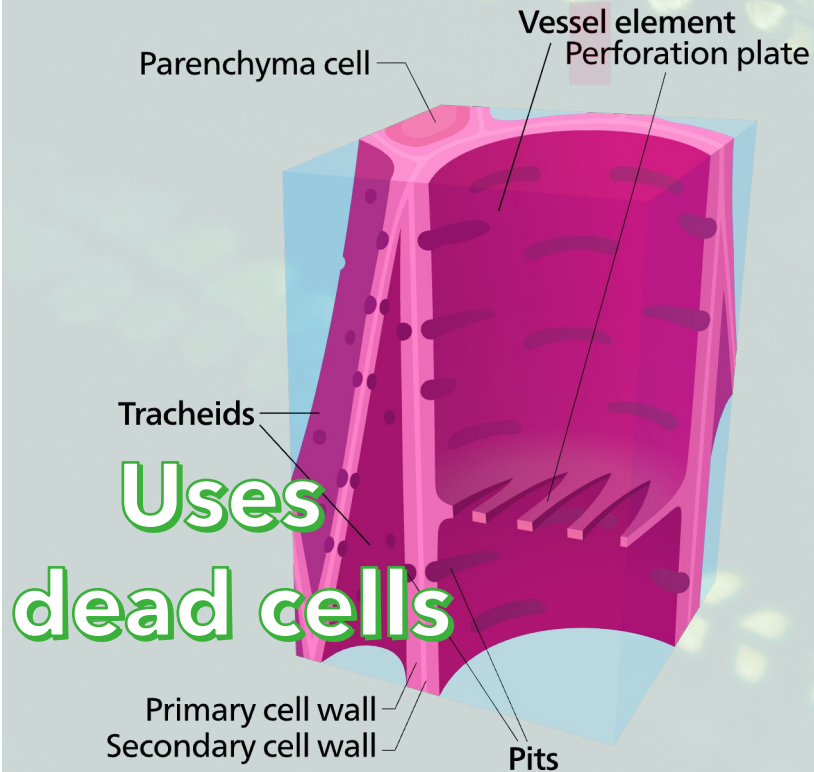
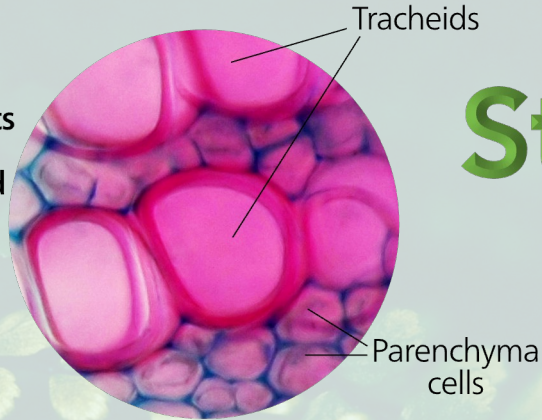
How Do They Work?

Cross Section of a Stem



Xylem

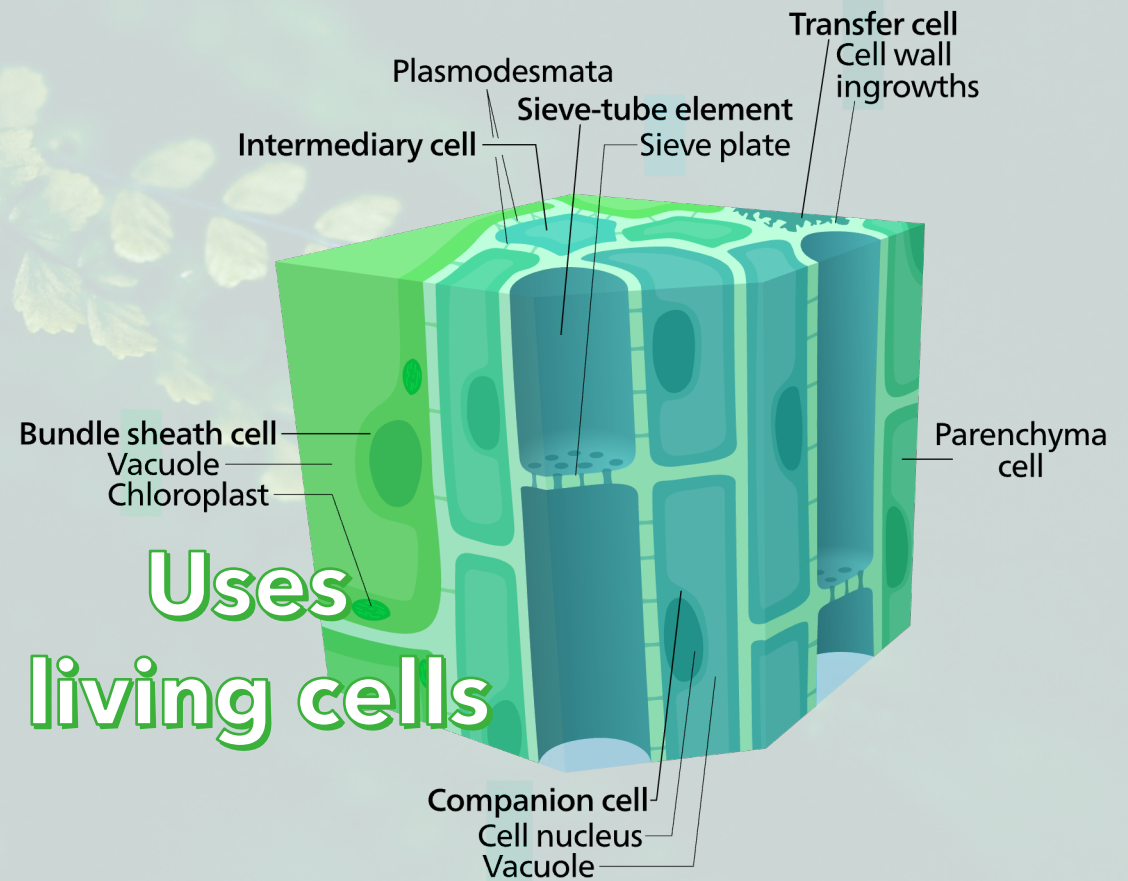
Xylem transports water and minerals through vessel elements and tracheids, which are dead at maturity and have a primary and secondary cell wall. In pits, the secondary wall is thin or missing, allowing water to flow laterally.



Structure

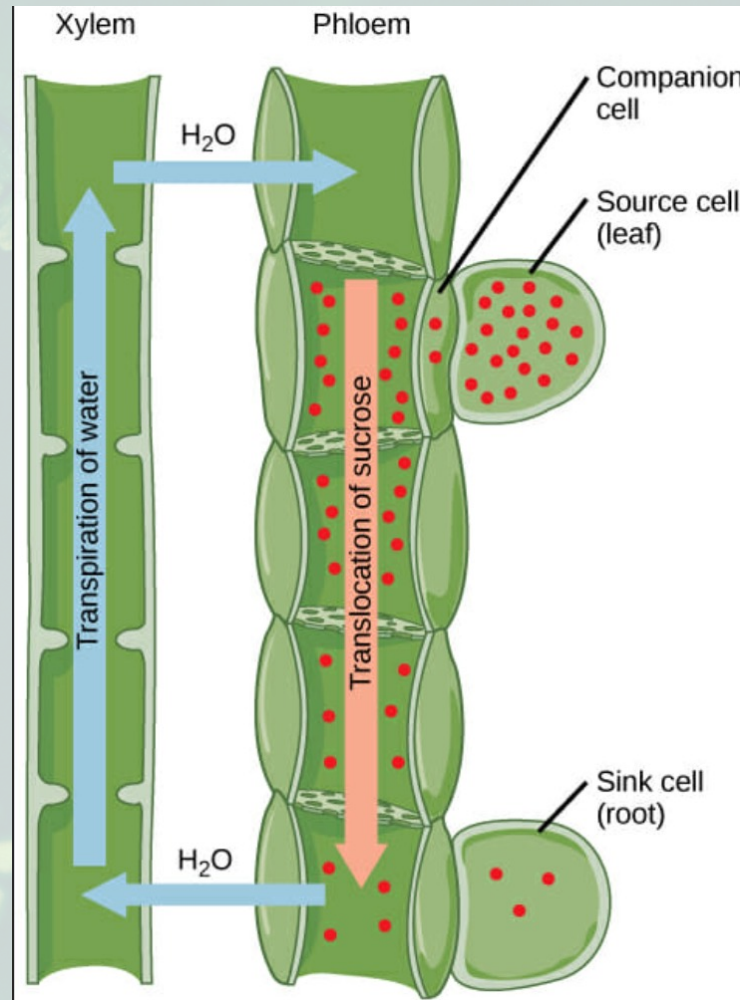
Phloem

Phloem transports sugars and other items. In angiosperms, sieve-tube elements contain the sugar solution. Sieve-tube cells are surrounded by various support cells.



How Do They Work?

Water is pulled up the xylem because it can cling to the side of the tube and the evaporating water from the leaves (transpiration) makes a difference in pressure



Some water from the xylem can flow into the phloem, which mixes with the sugars to make sap. The thinner sap can flow to other parts of the plant and be stored in the roots.

Examples of Vascular Plants



Club Moss (not a moss)



Ferns

Examples of Vascular Plants



Gymnosperms(exposed seed)



Angiosperms

Why Have Vascular Tissue?

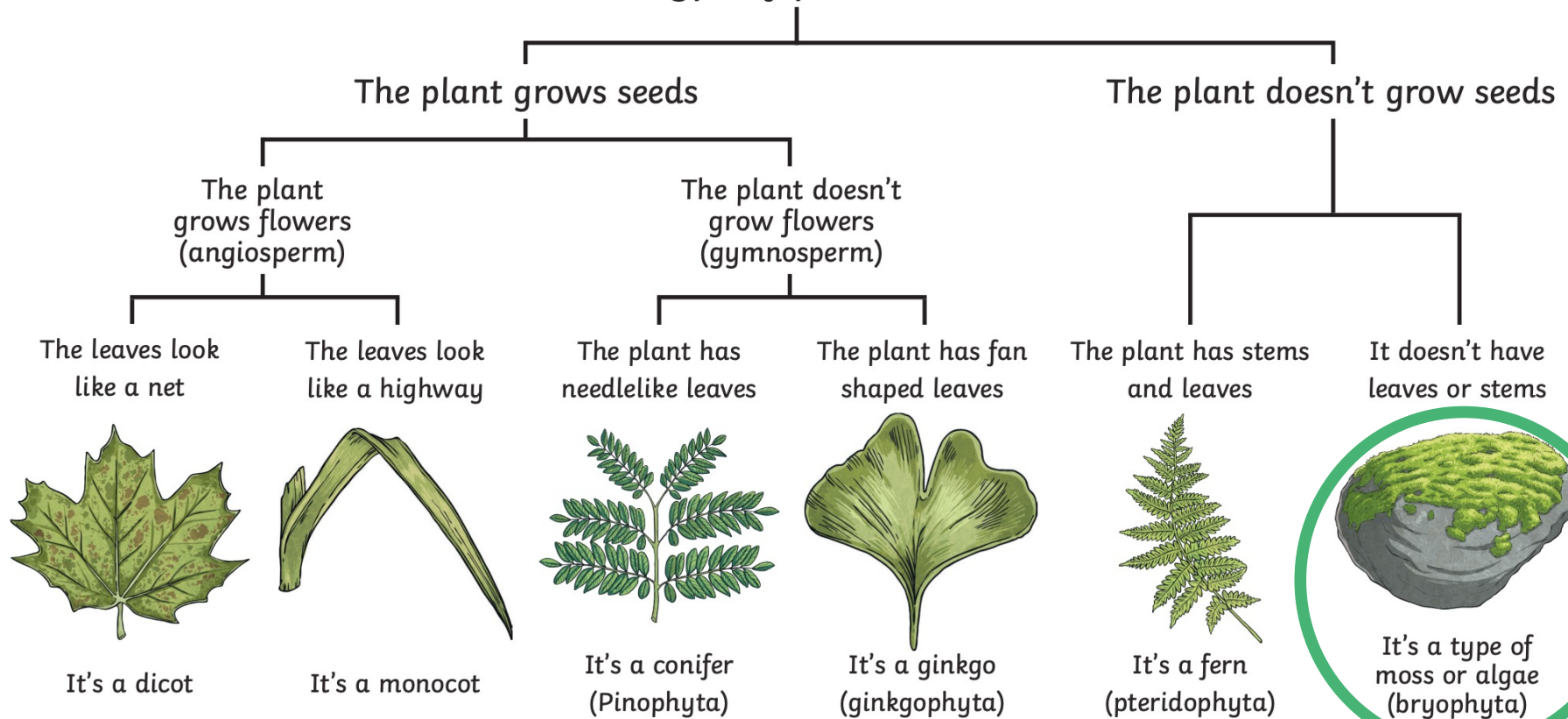
- Can grow far away from water sources
- Can support tall growth
- Can develop many new types of structures, such as wood, tubers, and flowers



Which are non-vascular?

Dichotomous Key for Plants

What type of plant is that?



We Have Learned

- Scientists divide land plants into vascular and non-vascular
- Vascular plants transport materials using special structures
- Xylem transport water up from the roots
- Phloem transport sugar from the leaves to the rest of the plant
- Mosses, liverwort and hornwort are nonvascular plants
- All ferns, gymnosperms, and angiosperms are vascular plants